🏥 Apteka 999 — Dorixona Yetkazib Berish Platformasi

Loyiha hujjati · Aprel 2026

📋 Mundarija

1. Loyiha haqida
2. Tizim arxitekturasi
3. Texnologiyalar
4. Modul va funksiyalar
5. Ma'lumotlar bazasi
6. API hujjati
7. Xavfsizlik
8. Deploy va infratuzilma
9. Loyiha narxi

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1. Loyiha haqida

Apteka 999 — Toshkent shahridagi 20 ta dorixona filialini birlashtirgan to'liq raqamli yetkazib berish platformasi. Platforma mijozlarga dori-darmonlarni qulay tarzda qidirish, buyurtma berish va to'lov qilish imkonini beradi.

Asosiy maqsadlar:

- 20 ta dorixona filialini yagona raqamli platformaga birlashtirish
- Telegram Mini App orqali foydalanuvchilarga qulay interfeys taqdim etish
- Buyurtma → To'lov → Yetkazib berish jarayonini to'liq avtomatlashtirish

- Real-time analitika va boshqaruv paneli orqali operatsion samaradorlikni oshirish
- Bonus tizimi orqali mijoz sadoqatini shakllantirish

Maqsadli auditoriya:

- **Mijozlar**: Toshkent aholisi, dorixona xizmatlari foydalanuvchilari
- **Operatorlar**: Filial xodimlari, buyurtma boshqaruvchilari
- **Kurierlar**: Yetkazib berish xodimlari
- **Administratorlar**: Tizim boshqaruvchilari

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2. Tizim Arxitekturasi

2.1 Umumiy arxitektura

```
┌──────────────────────────────────────────────────────────────────┐
│                          FOYDALANUVCHILAR                        │
│                                                                  │
│   📱 Telegram Mini App        🖥️ Admin Panel                     │
│   (React + Vite)              (React + Vite + Recharts)          │
│   Port: 5173                  Port: 5174                         │
└──────────────┬──────────────────────────────┬──────────────────┘
               │            HTTPS             │
               ▼                              ▼
┌──────────────────────────────────────────────────────────────────┐
│                      🔧 BACKEND API SERVER                       │
│                     Node.js + Express                            │
│                     Port: 3000                                   │
│                                                                  │
│  ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────┐            │
│  │   Auth   │ │ Orders   │ │ Products │ │ Payment  │            │
│  │ Routes   │ │ Routes   │ │ Routes   │ │ Routes   │            │
│  └──────────┘ └──────────┘ └──────────┘ └──────────┘            │
│                                                                  │
│  ┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────┐            │
│  │ Branches │ │Analytics │ │ Import   │ │ Admin    │            │
│  │ Routes   │ │ Routes   │ │ Routes   │ │ Routes   │            │
│  └──────────┘ └──────────┘ └──────────┘ └──────────┘            │
│                                                                  │
│  ┌──────────────────── SERVICES ────────────────────────────┐   │
│  │ Click · Payme · Bonus · Telegram · Analytics             │   │
│  │ Import · Aslbelgisi (Verification)                       │   │
│  └──────────────────────────────────────────────────────────┘   │
└──────────────┬──────────────────────────────┬──────────────────┘
               │                              │
               ▼                              ▼
┌──────────────────────┐   ┌──────────────────────────┐
│ 🍃 MongoDB 7         │   │ 🤖 Telegram Bot          │
│ Port: 27017          │   │ Grammy Framework         │
│                      │   │ Port: 3001               │
│ Collections:         │   │                          │
│ - users              │   │ Handlers:                │
│ - products           │   │ - start                  │
│ - orders             │   │ - operator               │
```

```
│  - branches        │    │  - courier          │
│  - stocks          │    │  - customer         │
│  - analytics       │    │                     │
│  - bonustx         │    │  Keyboards:         │
│  - settings        │    │  - inline menus     │
│  - adminaccount    │    │                     │
│  - importlogs      │    │  Locales:           │
└────────────────────┘    │  - uz, ru, en       │
                          └─────────────────────┘
```

2.2 Microservice Kommunikatsiya

```
                          Webhook
┌──────────────┐  ◄──────────────►  ┌──────────────┐
│  Telegram    │                    │  Bot Server  │
│  Server      │                    │  (port 3001) │
└──────────────┘                    └──────┬───────┘
                                           │
                                      HTTP Proxy
                                           │
                          REST API         ▼
┌──────────────┐  ◄──────────────►  ┌──────────────┐
│   Click      │                    │   Backend    │
│   Server     │   /click/*         │  (port 3000) │
└──────────────┘                    │              │
                                    │              │
                          JSON-RPC  │              │
┌──────────────┐  ◄──────────────►  │              │
│   Payme      │                    │              │
│   Server     │  /api/payment      │              │
└──────────────┘     /payme         └──────────────┘
```

2.3 Buyurtma Hayot Tsikli (Order Lifecycle)

```
                  ┌──────────────────┐
                  │ awaiting_payment │◄── Buyurtma yaratildi
                  └──────────┬───────┘
                             │ To'lov muvaffaqiyatli
                             ▼
                  ┌──────────────────┐
                  │ pending_operator │◄── Operator kutmoqda
                  └──────────┬───────┘
                            / \
              Tasdiqladi   Rad etdi
                          /       \
```

```
                    ▼              ▼
        ┌──────────┐   ┌──────────┐
        │ confirmed │   │ rejected │
        └──────────┘   └──────────┘
                 │
                 │ Kurier yo'lga chiqdi
                 ▼
        ┌──────────┐
        │ on_the_way│
        └──────────┘
                 │
                 │ Yetkazildi
                 ▼
        ┌──────────┐
        │ delivered │
        └──────────┘

    * Har qanday bosqichda → cancelled (bekor qilingan)
```

3. Texnologiyalar

3.1 Texnologiya Steki

Qatlam	Texnologiya	Versiya	Maqsad
Frontend (Mini App)	React	18.x	Telegram Mini App UI
Frontend (Admin)	React	18.x	Boshqaruv paneli
Build Tool	Vite	5.x	Tez build va HMR
Grafik	Recharts	2.x	Dashboard analitika
Backend	Node.js	20+	Server runtime
Framework	Express.js	4.x	REST API framework
Database	MongoDB	7.x	NoSQL ma'lumotlar bazasi
ODM	Mongoose	8.x	MongoDB obyekt modeli
Bot	Grammy	1.x	Telegram Bot API
Auth	JWT	—	JSON Web Token autentifikatsiya
Security	Helmet	—	HTTP xavfsizlik headerlari
Rate Limit	express-rate-limit	—	So'rovlarni cheklash
Container	Docker + Compose	—	Deployment
Hosting	Railway	—	Cloud PaaS

3.2 Tashqi integratsiyalar

Xizmat	Maqsad	Protokol
Click	Onlayn to'lov	REST (Prepare/Complete)
Payme	Onlayn to'lov	JSON-RPC 2.0

Telegram Bot API	Xabarnomalar, bot interaksiya	Webhook
Telegram WebApp API	Mini App autentifikatsiya	initData validation
Yandex Navigator	Kurier navigatsiya	Deep linking

4. Modul va Funksiyalar

4.1 📱 Telegram Mini App (Mijoz interfeysi)

Sahifa	Funksiya
Home	Asosiy sahifa, kategoriyalar, mashhur mahsulotlar
Catalog	Mahsulotlar katalogi, qidiruv, filtr
Product Detail	Mahsulot haqida batafsil ma'lumot, analog dorilar
Branches	20 ta filial ro'yxati, xarita, masofa bo'yicha saralash
Cart	Savatcha, miqdor boshqaruvi, bonus qo'llash
Payment	Click yoki Payme orqali to'lov
Scanner	Barkod skaneri orqali dori qidirish
Profile	Profil, buyurtmalar tarixi, manzillar, bonus

Maxsus xususiyatlar:

- 🔍 Latin → Kirill transliteratsiya qidiruvi (lotinchada yozib kirill natija topish)
- 📍 GPS orqali eng yaqin filial aniqlash
- 🎯 Telegram WebApp API to'liq integratsiya (haptic feedback, theme sync)
- 🌐 3 tilda interfeys (O'zbekcha, Ruscha, Inglizcha)

4.2 🖥️ Admin Panel (Boshqaruv paneli)

Sahifa	Funksiya
Dashboard	Real-time KPI, grafik, analitika, buyurtma funnel
Orders	Buyurtmalar ro'yxati, filtrlash, status boshqarish
Products	Mahsulot CRUD, kategoriya boshqarish

Branches	Filiallar boshqaruvi, operator tayinlash
Users	Foydalanuvchilar ro'yxati, bonus boshqarish
Import	Excel orqali ommaviy mahsulot yuklash
Accounts	Admin hisoblar boshqaruvi
Settings	Tizim sozlamalari (yetkazib berish narxi, bonus qoidalari)

Dashboard xususiyatlari:

- 📊 Animatsiyali KPI hisoblagichlar
- 📈 Gradient AreaChart grafiklar (Recharts)
- 🔮 Glassmorphism dizayn elementlari
- 💀 Skeleton loading holatlari
- 🔄 Buyurtma funnel vizualizatsiya
- ⏱️ So'nggi buyurtmalar real-time ro'yxati

4.3 🤖 Telegram Bot

Handler	Funksiya
Start	Foydalanuvchi ro'yxatdan o'tishi, til tanlash
Operator	Yangi buyurtma xabarnomasi, tasdiqlash/rad etish
Courier	Buyurtma olish, yetkazib berish tasdiqlash
Customer	Buyurtma holati xabarlari, bonus xabarlari

Bot xususiyatlari:

- 📩 Avtomatik buyurtma xabarnomasi operatorlarga
- ✅ Inline tasdiqlash/rad etish tugmalari
- 🚗 Kurierga buyurtma tayinlash
- 📍 Yandex Navigator orqali manzilga yo'naltirish

- 🌐 Ko'p tilli (uz, ru, en)
- 🛡️ Blocked bot graceful error handling

4.4 💳 To'lov Tizimi

Click integratsiyasi

Bosqich	Tavsif
Prepare	Buyurtma tekshiriladi, `merchant_prepare_id` qaytariladi
Complete	To'lov tasdiqlanadi, buyurtma statusi yangilanadi

Payme integratsiyasi (JSON-RPC 2.0)

Metod	Tavsif
`CheckPerformTransaction`	Buyurtma to'lov qilsa bo'ladimi tekshirish
`CreateTransaction`	Yangi tranzaksiya yaratish
`PerformTransaction`	To'lovni amalga oshirish
`CancelTransaction`	Tranzaksiyani bekor qilish
`CheckTransaction`	Tranzaksiya holatini tekshirish
`GetStatement`	Tranzaksiyalar hisoboti

To'lov xavfsizligi:

- ✅ Idempotent webhook qayta ishlash
- ✅ Timeout boshqaruvi (12 soat)
- ✅ State machine (5 holat: new → created → performed → cancelled)
- ✅ Frontend polling orqali status tekshirish

4.5 ⭐ Bonus Tizimi

Daraja	Ball oralig'i	Belgisi
Kumush	0 — 1,999 ball	🥈
Oltin	2,000 — 4,999 ball	🥇
Platina	5,000+ ball	💎

Qoidalar:

- Har 10,000 so'm xaridga → 100 bonus ball
- 1 ball = 1 so'm chegirma
- Maksimal chegirma: buyurtma summasining 30%

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5. Ma'lumotlar Bazasi

5.1 Kolleksiyalar

Kolleksiya	Maydonlar soni	Tavsif
users	15+	Telegram foydalanuvchilar, bonus, manzillar
products	12+	Dorilar katalogi, barcode, analog, tavsif (uz/ru)
orders	30+	Buyurtmalar, to'lov ma'lumoti, status tarixi
branches	10+	Filiallar, lokatsiya, operator/kurier ID
stocks	4+	Mahsulot zaxirasi (filial → mahsulot → narx → miqdor)
analyticsdailies	10+	Kunlik statistika agregatsiya
bonustransactions	6+	Bonus ball tarixi
settings	5+	Tizim sozlamalari
adminaccounts	4+	Admin hisoblar
importlogs	5+	Excel import tarixi

5.2 Asosiy Indekslar

```
orders   → { telegramId: 1 }, { createdAt: -1 }, { branch: 1, status: 1 }
products → { name: 'text', ingredient: 'text', manufacturer: 'text' }
users    → { telegramId: 1 } (unique)
branches → { number: 1 }
```

6. API Hujjati

6.1 Autentifikatsiya

Metod	Endpoint	Tavsif
`POST`	`/api/auth/telegram`	Telegram WebApp initData orqali auth
`POST`	`/api/auth/admin/login`	Admin panel login (JWT)

6.2 Mahsulotlar

Metod	Endpoint	Tavsif
`GET`	`/api/products`	Barcha mahsulotlar (pagination, search, filter)
`GET`	`/api/products/:id`	Bitta mahsulot
`POST`	`/api/products`	Yangi mahsulot (Admin)
`PUT`	`/api/products/:id`	Mahsulot tahrirlash (Admin)
`DELETE`	`/api/products/:id`	Mahsulot o'chirish (Admin)

6.3 Filiallar

Metod	Endpoint	Tavsif
`GET`	`/api/branches`	Barcha filiallar
`GET`	`/api/branches/:id`	Bitta filial
`GET`	`/api/branches/:id/stock`	Filial zahirasi
`POST`	`/api/branches`	Yangi filial (Admin)
`PUT`	`/api/branches/:id`	Filial tahrirlash (Admin)

6.4 Buyurtmalar

Metod	Endpoint	Tavsif
`POST`	`/api/orders`	Yangi buyurtma
`GET`	`/api/orders`	Foydalanuvchi buyurtmalari
`GET`	`/api/orders/:id`	Buyurtma tafsiloti
`PATCH`	`/api/orders/:id/status`	Status o'zgartirish (Admin)

6.5 To'lov

Metod	Endpoint	Tavsif
`POST`	`/api/payment/click/prepare`	Click prepare webhook
`POST`	`/api/payment/click/complete`	Click complete webhook
`POST`	`/api/payment/payme`	Payme JSON-RPC endpoint
`GET`	`/api/payment/status/:orderId`	To'lov holati tekshirish

6.6 Analitika

Metod	Endpoint	Tavsif
`GET`	`/api/analytics/dashboard`	Dashboard statistika
`GET`	`/api/analytics/daily`	Kunlik grafiklar ma'lumoti

6.7 Import

Metod	Endpoint	Tavsif
`POST`	`/api/import/excel`	Excel fayldan mahsulot import

6.8 Health Check

Metod	Endpoint	Tavsif

`GET`	`/api/health`	Tizim salomatligi

7. Xavfsizlik

7.1 Autentifikatsiya va Avtorizatsiya

Qatlam	Mexanizm
Mini App	Telegram WebApp `initData` HMAC-SHA256 tekshiruv
Admin Panel	JWT token (alohida `ADMIN_JWT_SECRET`)
Bot	`secret_token` webhook tekshiruv
API	Role-based access control (user/admin/operator/courier)

7.2 Tarmoq xavfsizligi

Himoya	Vosita
HTTP Headers	Helmet.js (XSS, CSRF, Clickjacking)
Rate Limiting	200 so'rov / 15 daqiqa
CORS	Credentialed cross-origin ruxsat
CSP	Telegram domenlariga frame-ancestors ruxsat
Input Validation	Request body tekshiruv

7.3 To'lov xavfsizligi

- Click: `sign_string` MD5 hash tekshiruv
- Payme: Basic Auth + merchant key tekshiruv
- Idempotent webhook qayta ishlash
- Transaction timeout monitoring (12 soat)

8. Deploy va Infratuzilma

8.1 Docker Compose Arxitekturasi

Servis	Image	Port	Tavsif
`mongodb`	mongo:7	27017	Ma'lumotlar bazasi
`backend`	Custom	3000	API server
`frontend`	Custom	5173 → 80	Telegram Mini App
`admin`	Custom	5174 → 80	Admin Panel
`bot`	Custom	3001	Telegram Bot

8.2 Deploy jarayoni

```
# 1. Repository klonlash
git clone <repository-url>

# 2. Environment sozlash
cp .env.example backend/.env
# BOT_TOKEN, MONGODB_URI, JWT_SECRET, ADMIN_JWT_SECRET → tahrirlang

# 3. Docker bilan ishga tushirish
docker-compose up -d

# 4. Yoki alohida ishga tushirish
cd backend && npm install && npm start
cd admin && npm install && npm run dev
cd frontend && npm install && npm run dev
cd bot && npm install && npm start
```

8.3 Cloud Deploy (Railway)

Loyiha Railway PaaS platformasida hosting qilinadi:

- Avtomatik CI/CD (git push → deploy)
- MongoDB Atlas ulash
- SSL sertifikat (avtomatik)
- Custom domain sozlash

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9. 💰 Loyiha Narxi

9.1 Ishlanma Tarkibi va Baholash

#	Modul	Murakkablik	Soat	Narx (so'm)
1	**Backend API Server**	Yuqori	120	18,000,000
	— Express.js arxitektura, middleware			
	— 11 ta route moduli			
	— 7 ta servis (to'lov, analitika, bonus)			
	— 10 ta MongoDB model			
2	**Telegram Mini App**	Yuqori	100	15,000,000
	— 8 ta sahifa (Home, Catalog, Cart, Payment...)			
	— Telegram WebApp API integratsiya			
	— 3 tilda interfeys (uz, ru, en)			
	— Responsive dizayn			
3	**Admin Panel**	Yuqori	90	13,500,000
	— Dashboard (Recharts, KPI, analitika)			
	— 7 ta boshqaruv sahifasi			
	— Enterprise-grade UI/UX			
4	**Telegram Bot**	O'rta	50	7,500,000
	— Grammy framework			
	— Operator/Courier/Customer handlers			
	— Ko'p tilli (3 til)			
5	**Click to'lov integratsiya**	Yuqori	40	6,000,000

	— Prepare/Complete webhook			
	— Xavfsizlik, idempotent qayta ishlash			
6	**Payme to'lov integratsiya**	Yuqori	45	6,750,000
	— JSON-RPC 2.0 to'liq integratsiya			
	— 6 ta metod, state machine			
7	**Bonus tizimi**	O'rta	25	3,750,000
	— 3 darajali tizim			
	— Avtomatik hisoblash			
8	**Excel Import moduli**	O'rta	20	3,000,000
	— Bulk mahsulot yuklash			
	— Import loglari			
9	**Docker + Deploy**	O'rta	20	3,000,000
	— Docker Compose (5 servis)			
	— Railway CI/CD			
10	**Dizayn va UX**	O'rta	30	4,500,000
	— Modern glassmorphism dizayn			
	— Responsive, animatsiyalar			

9.2 Umumiy narx xulosasi

Jami ish soatlari	**540 soat**
Modullar soni	**10 ta**
Kodlar bazasi	**~50 ta fayl, 15,000+ qator**

Taqdimot	Narx
💡 **Umumiy loyiha narxi**	**81,000,000 so'm**
🔧 Texnik qo'llab-quvvatlash (6 oy)	5,000,000 so'm
📚 Hujjatlashtirish	Bepul
📦 **TO'LIQ PAKET NARXI**	**86,000,000 so'm**

> **Eslatma**: Narxlar ishlab chiquvchining tajribasi, loyihaning murakkabligi va bozor narxlariga asosan belgilanadi. Narxlar kelishuv asosida o'zgarishi mumkin.

9.3 Qo'shimcha xizmatlar (ixtiyoriy)

Xizmat	Narx
Yangi to'lov tizimi integratsiya	5,000,000 so'm
Mobil ilova (React Native)	30,000,000 so'm
SMS xabar integratsiya	3,000,000 so'm
Qo'shimcha analitika moduli	5,000,000 so'm
Server monitoring (Grafana)	4,000,000 so'm

📞 Aloqa

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